Table 1 — Genomics, Molecular Biology and Life Sciences

BENQODHI Day 2 morning round tables · 11 September 2026

Printable guide for the table. Keep it in front of you during the discussion.

Aim of the round tables

Collect focused expert input for a possible BENQODHI output (paper, report, roadmap or policy note). The central question is how important challenges in health and the life sciences can become **credible, reusable benchmarks** for classical, AI-based, hybrid and quantum algorithms.

The preferred focus is **optimization** problems, but the discussion is not restricted to them: inference, sampling, simulation, reconstruction and learning problems also count when they have a clear health or life-science motivation and a plausible route to benchmarkability.

We are **not** building the benchmark repository during the workshop. The goal is to identify which problems would make a future benchmark repository useful.

The discussion path

> **health or life-science challenge → computational bottleneck → candidate benchmark → available evidence and missing pieces → fair comparison of algorithms → next benchmark-building action**

This table's role

Identify candidate genomics and life-science problems that could become benchmarkable. First collect possible candidates (from the prompts below, the problem brief, or your own
expertise), then select the strongest ones and answer the questions for each.

Problem prompts for discussion

Starting points only — replace or add to them when stronger candidates emerge.

- haplotype phasing and related variant-interpretation problems;
- RNA/DNA sequence similarity, matching, alignment or structure prediction;
- phylogenetic, hierarchical clustering or MQTC-like tree reconstruction problems;
- genomic benchmark-data standards, truth sets and difficult genomic regions;
- molecular or biological-network problems with a clear computational task and domain-relevant metric.

Focus: identify which life-science problems are important, computationally precise and realistic enough to assess for benchmarkability.

Questions to answer

For each selected candidate problem (aim for 2–3 candidates)

1. **Problem** — What health or life-science problem should be benchmarked, and why does it matter?
2. **Bottleneck** — What is hard computationally?
3. **Moving ahead** — What is missing, and what is the next practical step?

Transversal question — answer once for the table

Sustaining the ecosystem (infrastructure, community, scientific governance): BENQODHI only works if what this table produces feeds one connected, durable ecosystem rather than a silo. Of the three pillars — the shared **infrastructure** that hosts and maintains it, the **community** that keeps contributing, and the **scientific governance** that keeps it credible — how can this table best feed the third pillar, scientific governance? What would it take for the problems, data and baselines from this table to be curated, reviewed and trusted as credible science over time, and what governance role should this table's own community play in setting and upholding those standards?

Expected output

- a short list of selected candidate problems;
- for each candidate, short answers to the three questions above;
- one answer to the transversal governance question.

Note takers: draft answers live into `answers.md` in this folder. Rapporteur: present the main conclusion, main candidate problem, why it matters, the most important missing piece, the next action, and the governance answer.